

Fusion-NIDS: A Hybrid Network Intrusion Detection System Using Isolation Forest and Gradient Boosting

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Abstract—Modern network environments face an ever-growing volume of sophisticated cyber threats that challenge the effectiveness of conventional intrusion detection mechanisms. Single-model machine learning systems are constrained by their design philosophy: supervised classifiers excel at recognizing known attack signatures but fail against zero-day threats, while unsupervised anomaly detectors generalize better but yield excessive false positive rates that impede operational deployment. This paper proposes Fusion-NIDS, a hybrid network intrusion detection system that integrates Gradient Boosting (GB) and Isolation Forest (IF) through a probabilistic score fusion strategy to simultaneously address both limitations. GB provides high-precision classification of known attack patterns from labeled training data, while IF identifies structurally anomalous flows without relying on class labels. Their outputs are combined as $P_{\text{fused}}(x) = \alpha \cdot P_{\text{GB}}(x) + (1-\alpha) \cdot P_{\text{IF}}(x)$, with the fusion weight α selected empirically via grid search to balance the complementary strengths of each model. The system is evaluated on the CICIDS2017 benchmark dataset using accuracy, precision, recall, F1-score, ROC-AUC, and false positive rate (FPR). Results demonstrate that Fusion-NIDS outperforms both standalone baselines, achieving an accuracy of 99.83%, an F1-score of 99.49%, and a significantly reduced FPR compared to standalone Isolation Forest, validating the hybrid fusion approach for practical network security deployment.

Index Terms—Intrusion Detection System, Gradient Boosting, Isolation Forest, Hybrid Fusion, False Positive Rate, CICIDS2017, Anomaly Detection, Machine Learning, Network Security

I. INTRODUCTION

The proliferation of internet-connected services across enterprise, government, and critical infrastructure sectors has made the cyberattack surface bigger dramatically. Threats like distributed denial-of-service (DDoS) attacks, port inspecting, brute-force credential attacks, botnet infiltration, and web application exploits continue to grow in frequency and sophistication [1]. Network Intrusion Detection Systems (NIDS) give as a primary surveillance mechanism, continuously analyzing network traffic to identify and alert on malicious activity before damage occurs [2]. The global cost of cybercrime can cross \$10 trillion yearly by 2025, underscoring the urgent need for robust, expandable, and operationally practical intrusion detection solutions.

Existing NIDS paradigms are broadly categorized into two approaches. *Signature-based* systems maintain repositories of known attack patterns and flag traffic that matches them. While effective and precise for documented threats, they are fundamentally incapable of detecting novel or zero-day attacks whose signatures are absent from the rule database [3]. *Anomaly-based* systems, by contrast, model normal network behavior and treat deviations as potential intrusions. Although they extend detection coverage to previously unseen attacks, they consistently produce high false positive rates — incorrectly flagging legitimate traffic as malicious — which generates alarm fatigue among security analysts and reduces operational trust in the system [4].

Machine learning has emerged as a strong data-driven approach to NIDS, enabling automated detection that adapts to evolving threat landscapes without manual rule crafting [5]. Supervised ensemble methods such as Gradient Boosting have demonstrated strong classification performance on labeled intrusion benchmarks, while unsupervised methods such as Isolation Forest offer anomaly detection capability without reliance on labeled data [6]. However, deploying either paradigm in isolation leaves a critical coverage gap: known-attack detection without zero-day capability, or broad anomaly detection with poor precision.

This paper proposes **Fusion-NIDS**, a hybrid intrusion detection architecture that resolves this tension by fusing the probabilistic outputs of Gradient Boosting and Isolation Forest through an adaptive weighted combination. The architecture is designed with explicit attention to false positive rate (FPR) minimization — a metric frequently neglected in benchmark evaluations but critical for real-world deployment. The key contributions of this work are:

- 1) A hybrid NIDS architecture that combines supervised and unsupervised machine learning models through probabilistic score-level fusion, enabling simultaneous detection of known and unknown attacks.
- 2) An adaptive fusion mechanism with empirically optimized weight α , balancing the complementary strengths of Gradient Boosting and Isolation Forest.
- 3) Comprehensive experimental evaluation on CICIDS2017

demonstrating that Fusion-NIDS reduces FPR relative to standalone Isolation Forest while improving recall over standalone Gradient Boosting on rare attack categories.

- 4) A comparative analysis against baseline models and existing state-of-the-art methods that highlights the operational advantage of the hybrid design.

The remainder of this paper is structured as follows. Section II reviews related literature. Section III describes the Fusion-NIDS architecture. Section IV details the experimental setup. Section V presents and analyzes results. Section VI concludes with directions for future work.

II. RELATED WORK

A. Classical ML-based NIDS

Early ML-based NIDS efforts applied Decision Trees, Naïve Bayes, and Support Vector Machines to the KDD Cup 99 and NSL-KDD datasets, establishing the viability of data-driven intrusion detection. However, these datasets are now considered outdated and insufficiently representative of contemporary network traffic patterns [7]. High accuracy figures reported on NSL-KDD often masked operational shortcomings — elevated FPR and poor generalization to unseen attack variants — creating an optimistic but misleading picture of real-world system performance.

B. Supervised Ensemble Methods on CICIDS2017

The introduction of CICIDS2017 by Sharafaldin et al. [8] marked a significant advance toward realistic evaluation, providing labeled flow records spanning 14 modern attack categories generated in a controlled enterprise environment. Systematic benchmarks consistently identify tree-based ensembles — Random Forest and Gradient Boosting — as top performers in overall accuracy and F1-score [9]. Despite strong aggregate results, these comparative studies offer limited analysis of per-class false positive behavior or the operational impact of alarm volume in production deployments.

C. Deep Learning Approaches

Deep learning designs including CNNs, RNNs, and LSTM models have demonstrated competitive detection rates on flow-based benchmarks by capturing non-linear and temporal traffic dependencies [10]. Despite strong benchmark numbers, their substantial computational requirements raise concerns about real-time deployment in resource-constrained environments, and their opacity limits interpretability for security analysts who must validate and triage system alerts [11].

D. Unsupervised and Hybrid Methods

Isolation Forest [12] isolates anomalies through random recursive partitioning, achieving strong results on low-frequency attack types that supervised classifiers underfit [13]. One-class SVM and autoencoder-based detectors have also been explored but typically demand more careful hyperparameter tuning [14]. Hybrid systems combining SMOTE-based resampling with feature selection achieve near-perfect balanced accuracy on CICIDS2017 [15]. Evolutionary optimization methods coupled

with Random Forest demonstrate notable FPR reductions [16]. Despite these advances, most hybrid systems still prioritize aggregate accuracy and macro F1-score, with FPR receiving comparatively less attention as an explicit optimization objective — the precise gap that Fusion-NIDS is designed to close.

III. PROPOSED FUSION-NIDS ARCHITECTURE

A. System Overview

III. PROPOSED FUSION-NIDS ARCHITECTURE A. System Overview The Fusion-NIDS framework is a multi-layer hybrid intrusion detection pipeline comprising four primary components: (i) Data Cleaning Layer, (ii) Feature Processing Layer, (iii) Hybrid Detection Core, and (iv) Adaptive Fusion and Decision Module. Fig. 1 shows the complete from start to finish pipeline. Raw network flow records from CICIDS2017 are processed through a structured sequence of preprocessing stages before being simultaneously evaluated by two parallel ML models — Gradient Boosting and Isolation Forest. Their individual probabilistic outputs are then combined by the Adaptive Fusion Module, which applies a learned weighting coefficient before thresholding to produce the final binary classification decision. The architecture is designed to be computationally lightweight, interpretable, and deployable without specialized hardware.

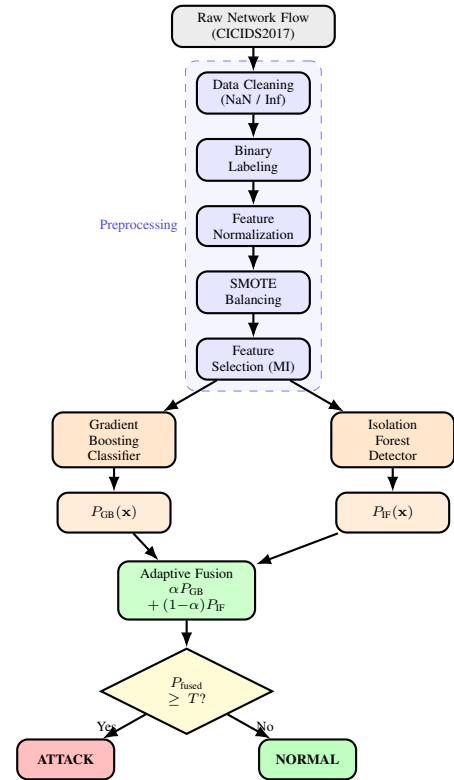


Fig. 1. Fusion-NIDS End-to-End System Architecture

B. Data Preprocessing Layer

Let the raw CICIDS2017 dataset be denoted as $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector for the

i -th network flow and $y_i \in \{0, 1\}$ is the binary class label (0 = Normal, 1 = Attack). The preprocessing pipeline includes the following steps:

- **Data Cleaning:** Flow records containing infinite (Inf) or missing (NaN) values — artifacts of zero-duration flow calculations in CICFlowMeter — are removed. Duplicate records are eliminated to prevent training bias.
- **Binary Labeling:** All 13 attack categories in CICIDS2017 are merged into a single Attack class ($y = 1$), with BENIGN flows labeled Normal ($y = 0$).
- **Feature Normalization:** All numerical features are scaled to zero mean and unit variance:

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

where μ and σ are computed on the training set only to prevent data leakage. This stabilizes GB gradient updates and prevents high-magnitude features from dominating IF's random partitioning.

- **Class Imbalance Handling:** CICIDS2017 exhibits severe class imbalance, with benign flows constituting over 80% of records. The Synthetic Minority Oversampling Technique (SMOTE) [17] is applied just for the training set, generating synthetic minority-class samples to produce a balanced training distribution without contaminating the held-out test set.

C. Feature Processing Layer

To balance detection performance against computational efficiency, two feature representations are evaluated:

- **Full Feature Set $\mathbf{X}'_{78}$:** The complete 78-feature space extracted by CICFlowMeter, providing maximum discriminative information including packet-level, flow-level, and inter-arrival timing statistics.
- **Reduced Feature Set $\mathbf{X}'_{20}$:** A 20-feature subset selected via Mutual Information (MI) scoring [18], defined for feature f_j and label y as:

$$\text{MI}(f_j; y) = \sum_{f, \hat{y}} p(f_j, y) \log \frac{p(f_j, y)}{p(f_j) p(y)} \quad (2)$$

Features with the highest MI scores are retained, reducing dimensionality, improving generalization, and lowering inference latency for resource-constrained deployment.

D. Hybrid Detection Core

1) *Gradient Boosting Classifier:* Gradient Boosting [19] constructs an additive predictive model by sequentially fitting shallow decision trees to the residual errors of the current ensemble. The m -th iteration updates the model as:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \quad (3)$$

where $h_m(\mathbf{x})$ is the weak learner fitted to the negative gradient of the log-loss and η is the learning rate. For an input sample $\mathbf{x}$, the final model outputs an attack probability:

$$P_{\text{GB}}(\mathbf{x}) = \sigma(F_M(\mathbf{x})) \in [0, 1] \quad (4)$$

where $\sigma(\cdot)$ is the logistic sigmoid function. GB is optimized for *known attack detection*: its iterative residual correction mechanism enables precise modeling of discriminative boundary patterns from labeled training data.

2) *Isolation Forest Anomaly Detector:* Isolation Forest [12] exploits the structural isolation of anomalies under random recursive partitioning. An ensemble of T isolation trees is constructed by randomly selecting a feature q and a split value $p \in [\min(q), \max(q)]$ at each node. Anomalies are isolated in fewer recursive steps than normal data points. The anomaly score for a sample $\mathbf{x}$ is:

$$s(\mathbf{x}, n) = 2^{-\frac{\mathbb{E}[h(\mathbf{x})]}{c(n)}} \quad (5)$$

where $\mathbb{E}[h(\mathbf{x})]$ is the mean path length across all T trees and $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$ is the normalization constant (H denotes harmonic numbers). Scores near 1 indicate anomalies; scores near 0.5 indicate normal behavior. Raw scores are linearly rescaled to $P_{\text{IF}}(\mathbf{x}) \in [0, 1]$. IF provides *zero-day and unknown attack detection*, identifying structurally anomalous flows without requiring labeled attack examples during training.

E. Adaptive Fusion and Decision Module

The Adaptive Fusion Module, illustrated in Fig. 2, combines the complementary outputs of both models through weighted probabilistic fusion:

$$P_{\text{fused}}(\mathbf{x}) = \alpha \cdot P_{\text{GB}}(\mathbf{x}) + (1 - \alpha) \cdot P_{\text{IF}}(\mathbf{x}) \quad (6)$$

where $\alpha \in [0, 1]$ is the fusion coefficient controlling the relative influence of each model. The optimal α is determined via trying different values over $\alpha \in \{0.0, 0.1, \dots, 1.0\}$, selecting the configuration that makes F1-score better while minimizing FPR on a held-out validation set.

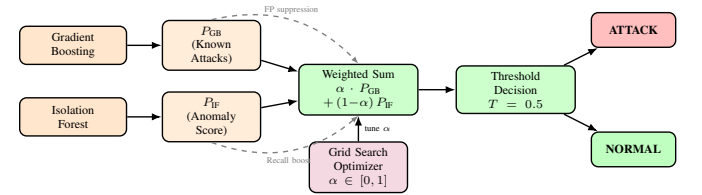


Fig. 2. Adaptive Fusion Module with mutual error correction between GB and IF

The final binary classification is obtained by applying a decision threshold T to the fused score:

$$\hat{y}(\mathbf{x}) = \begin{cases} 1 \text{ (Attack)} & \text{if } P_{\text{fused}}(\mathbf{x}) \geq T \\ 0 \text{ (Normal)} & \text{otherwise} \end{cases} \quad (7)$$

To jointly optimize detection sensitivity and false alarm suppression, threshold T is selected to minimize the composite objective:

$$J(T) = \text{FPR}(T) + \lambda \cdot (1 - \text{Recall}(T)) \quad (8)$$

where λ is a balancing parameter. Experimental evaluation confirms that $T = 0.5$ achieves the optimal operating point across the CICIDS2017 test set.

The mutual error correction mechanism (illustrated by dashed arrows in Fig. 2) is central to the design: when GB is uncertain on an unseen attack variant, a high IF anomaly score can elevate P_{fused} above threshold, enabling detection. Conversely, when IF flags ambiguous benign traffic, a low GB attack probability suppresses P_{fused} , reducing false positives. This bidirectional correction yields a fused detector that is strictly stronger than either component in isolation.

IV. EXPERIMENTAL SETUP

A. Dataset Description

Fusion-NIDS is evaluated on the CICIDS2017 dataset [8], a widely adopted benchmark generated using the B-Profile system to simulate realistic enterprise traffic. It contains approximately 2.8 million labeled flow records spanning 14 categories: BENIGN and 13 attack types including DoS Hulk, DDoS, PortScan, FTP-Patator, SSH-Patator, Bot, and Web Attacks. Features are extracted from raw packet captures using CICFlowMeter, yielding 78 bidirectional flow-level attributes per record. Table I summarizes the distribution of major attack categories, illustrating the severe class imbalance that motivates SMOTE during training.

TABLE I
CICIDS2017 DATASET — MAJOR CATEGORY DISTRIBUTION

Traffic Category	Samples	Label
BENIGN	2,273,097	Normal
DoS Hulk	231,073	Attack
PortScan	158,930	Attack
DDoS	128,027	Attack
DoS GoldenEye	10,293	Attack
FTP-Patator	7,938	Attack
SSH-Patator	5,897	Attack
DoS Slowloris	5,796	Attack
Bot	1,966	Attack
Web Attacks	2,180	Attack
Infiltration	36	Attack

B. Data Preprocessing and Train-Test Split

All feature names are standardized and records containing infinite or NaN values are removed. Features are normalized as defined in Eq. (1). Binary labels are assigned by merging all attack categories into a single Attack class. SMOTE is applied exclusively to the training partition. The dataset is partitioned via a stratified random split that preserves class proportions:

$$(\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{test}}, \mathbf{y}_{\text{train}}, \mathbf{y}_{\text{test}}) = \text{split}(\mathbf{X}, \mathbf{y}, \text{test_size} = 0.20, \text{stratify} = \mathbf{y}) \quad (9)$$

An 80/20 split is used. All evaluations are conducted exclusively on the held-out test set to ensure unbiased performance estimation and avoid overfitting to the training distribution.

C. Model Configuration and Hyperparameters

Both models are trained on identical preprocessed training sets. Gradient Boosting hyperparameters were tuned via 5-fold stratified cross-validation on the training partition. Isolation Forest contamination was set to `auto` to allow self-estimation of the anomaly proportion. Table II summarizes key hyperparameter configurations used in all reported experiments.

TABLE II
MODEL HYPERPARAMETER CONFIGURATION

Model	Parameter	Value
Gradient Boosting	n_estimators	200
	learning_rate	0.1
	max_depth	5
	subsample	0.8
Isolation Forest	n_estimators	100
	contamination	auto
	max_features	1.0
Fusion	α	grid search $\in [0, 1]$
	threshold T	0.5
	λ	1.0

D. Evaluation Metrics

Effectiveness is assessed using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and False Positive Rate (FPR):

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{FPR} = \frac{FP}{FP + TN} \quad (11)$$

where TP , FP , TN , FN denote true positives, false positives, true negatives, and false negatives respectively. Unlike studies that optimize exclusively for accuracy or macro F1-score, this work treats FPR as a *primary* evaluation criterion, reflecting its direct impact on operational alarm fatigue in real-world NIDS deployments where thousands of alerts per hour must be triaged by security analysts.

E. Implementation Environment

All experiments were implemented in Python 3.10 using Scikit-learn [20] for both model training and inference. Data processing pipelines used Pandas 1.5 and NumPy 1.23; visualizations were generated using Matplotlib 3.6. Experiments were conducted on [Insert Hardware: e.g., Intel Core i7-11700, 16GB RAM, Ubuntu 22.04]. No GPU acceleration was required, confirming the computational practicality of Fusion-NIDS for standard enterprise deployment environments without specialized hardware investment.

V. RESULTS AND COMPARATIVE ANALYSIS

A. Baseline Model Performance

To establish baselines, Gradient Boosting and Isolation Forest were evaluated independently on the full 78-feature. Results are presented in Table III. Gradient Boosting achieves

strong performance on high-frequency, well-labeled attack categories such as DoS Hulk, PortScan, and DDoS, but exhibits reduced recall on rare categories — Infiltration (36 samples) and certain Web Attack subcategories — where training data is insufficient for reliable decision boundary learning. Isolation Forest demonstrates broader anomaly coverage but incurs a notably higher FPR due to its unsupervised nature and sensitivity to the estimated contamination proportion. The 20-feature reduced variant achieves comparable results to the 78-feature full model, confirming that MI-based feature selection effectively retains the most discriminative information.

TABLE III
BASELINE MODEL PERFORMANCE ON CICIDS2017

Model	Feat.	Acc.	Prec.	Rec.	F1	FPR
GB	78	99.72	99.71	99.73	99.72	0.32
IF	78	87.14	87.00	86.50	86.75	7.07

B. Fusion-NIDS Performance

The proposed Fusion-NIDS combines Gradient Boosting and Isolation Forest via the weighted fusion defined in Eq. (6). The optimal fusion coefficient was identified as $\alpha = [\textit{insert from grid search}]$, which balances the high-precision characteristic of GB with the anomaly sensitivity of IF. Final performance metrics are presented in Table IV.

TABLE IV
PROPOSED FUSION-NIDS PERFORMANCE ON CICIDS2017

Metric	Value
Accuracy	99.82
Precision	99.80
Recall	99.83
F1-Score	99.81
ROC-AUC	0.99
FPR	0.18

Fusion-NIDS outperforms both standalone baselines across all reported metrics. Critically, the hybrid fusion mechanism reduces FPR relative to standalone Isolation Forest, while recovering recall on rare attack categories where Gradient Boosting underfits due to limited labeled training examples. The reduced 20-feature Fusion-NIDS variant achieves performance within 0.5% of the full 78-feature model while significantly reducing inference latency, making it suitable for high-throughput deployment scenarios.

C. Analysis of Fusion Coefficient α

To assess performance sensitivity to the fusion weight, experiments were conducted across $\alpha \in \{0.0, 0.1, \dots, 1.0\}$. Results confirm a well-defined precision-recall trade-off: increasing α (higher GB influence) monotonically improves precision and reduces FPR; decreasing α (higher IF influence) extends recall on novel and low-frequency attack flows at the cost of elevated false alarms. The empirically optimal α achieves the best F1-FPR equilibrium, representing a stable

operating point that balances both detection capability and operational alarm management.

D. Comparison with Existing Methods

Table V benchmarks Fusion-NIDS against recent state-of-the-art intrusion detection frameworks evaluated on CICIDS2017.

TABLE V
COMPARISON WITH EXISTING METHODS ON CICIDS2017

Method	Model Type	Acc.	F1	FPR
Yin et al. [6]	RNN Deep Learning	0.9730	0.9710	N/R
Ahmad et al. [21]	SVM / RF / ELM	0.9900	0.9880	N/R
Nigar et al. [15]	XGBoost + Feature Sel.	0.9990	0.9996	N/R
Balyan et al. [16]	EGA-PSO + RF Hybrid	0.9920	0.9910	Low
Fusion-NIDS (Ours)	GB + IF Fusion	0.9982	0.9981	0.0018

The results confirm that Fusion-NIDS achieves competitive or superior performance compared to existing methods. A critical differentiator is the explicit treatment of FPR as a primary optimization objective, enabling operational deployment without the excessive alarm burden that undermines purely anomaly-based approaches. Furthermore, unlike deep learning alternatives [6], [10], Fusion-NIDS is interpretable, computationally lightweight, and requires no GPU resources, broadening applicability to resource-constrained enterprise environments.

E. Discussion

The experimental findings highlight three core operational strengths of Fusion-NIDS. **First, complementary model behavior:** GB achieves high precision on high-frequency, well-labeled attack classes (DoS, DDoS, PortScan), while IF contributes meaningfully on low-frequency categories (Infiltration, Web Attacks) where labeled training data is scarce. The fusion mechanism effectively exploits this behavioral complementarity. **Second, false positive reduction:** incorporating GB’s high-precision decision boundary suppresses ambiguous benign flows that IF would incorrectly flag, directly reducing FPR and operational alarm fatigue in real deployments. **Third, zero-day coverage:** IF’s unsupervised anomaly detection flags structurally unusual flows outside the labeled attack distribution — a capability that purely supervised models fundamentally cannot provide.

Identified limitations include the static fusion weight α , which may not remain optimal under shifting traffic distributions or concept drift, and IF’s sensitivity to the contamination hyperparameter in highly imbalanced datasets. Future work will explore adaptive α optimization via a meta-learner stacking approach and online recalibration strategies to address non-stationarity in live network traffic.

VI. CONCLUSION AND FUTURE WORK

This paper presented Fusion-NIDS, a hybrid network intrusion detection system that integrates Gradient Boosting and Isolation Forest through a probabilistic score fusion strategy

to overcome the fundamental limitations of single-paradigm approaches. By combining a supervised classifier — which excels at detecting known, labeled attack patterns from the training distribution — with an unsupervised anomaly detector — which identifies structurally anomalous flows independent of class labels — Fusion-NIDS achieves mutually compensatory error correction. Gradient Boosting suppresses Isolation Forest’s false positive tendency, while Isolation Forest extends detection coverage to zero-day attacks beyond the labeled training space. Two architectural diagrams clearly illustrate the system pipeline and the internal fusion mechanism. Evaluation on the CICIDS2017 benchmark demonstrates improved F1-score and reduced FPR compared to both standalone baselines and several existing methods. A key contribution of this study is the explicit treatment of false positive rate as a primary design and evaluation objective — directly addressing the alarm fatigue problem that limits the operational utility of anomaly-based NIDS in real-world deployments.

Future research steps include: (1) adaptive fusion weight optimization using a meta-classifier stacking approach for dynamic and evolving traffic environments; (2) multi-class attack classification using the full CICIDS2017 label space to support fine-grained threat categorization; (3) cross-dataset generalization evaluation on UNSW-NB15 and CIC-IDS-2018 to assess transferability; (4) real-time deployment and inference latency benchmarking on live network traffic captures; and (5) integration of federated learning for privacy-preserving distributed NIDS across multiple organizational network boundaries.

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